

Appendix 8 How to Parse and Decipher the Onboard CAN Protocol - DBC Protocol

Instruction Manual

Instruction Manual Version: V2.04

Update Date: 2023.07.01

Automotive Private Protocols

(Other Industrial Equipment Private Protocols Are Subject to Similar Cracking)

The Automotive CAN bus data includes some standard protocols such as ISO15765 and J1939. However, the remaining CAN data is specific to the car manufacturer's private protocol, which is defined internally according to their own needs. This protocol is usually only known by the car manufacturer and is described in a PDF or DBC format file. Some front-end and aftermarket automotive equipment manufacturers collaborate with automotive original equipment manufacturers, and they have a corresponding agreement in place. For regular customers, such as manufacturers of automotive aftermarket products like navigation systems, decoders, reverse cameras, reverse radars, and 360 panorama cameras, it is necessary to obtain signals from the car's gear (mainly reverse), turn signals, double blinkers, handbrake, headlights, and other signals. This requires reading the original CAN data and analyzing it to develop their own solution.

In most cases, there is only one CAN bus on a car, except for German cars. This bus is connected to the CAN pin of the OBD port with a baud rate of 500K high-speed CAN. At this point, the CAN data can be directly read from the OBD port. For German cars like Volkswagen, there can be up to 5-6 CAN buses in the car. These include a fault-tolerant CAN with a baud rate of 100K and a high-speed CAN with a baud rate of 500K. The body data is not sent out through the OBD port due to the presence of a gateway. Therefore, it can only be read through the corresponding CAN line in the body. For example, the navigation system is connected to the comfortable CAN (100K fault-tolerant CAN).

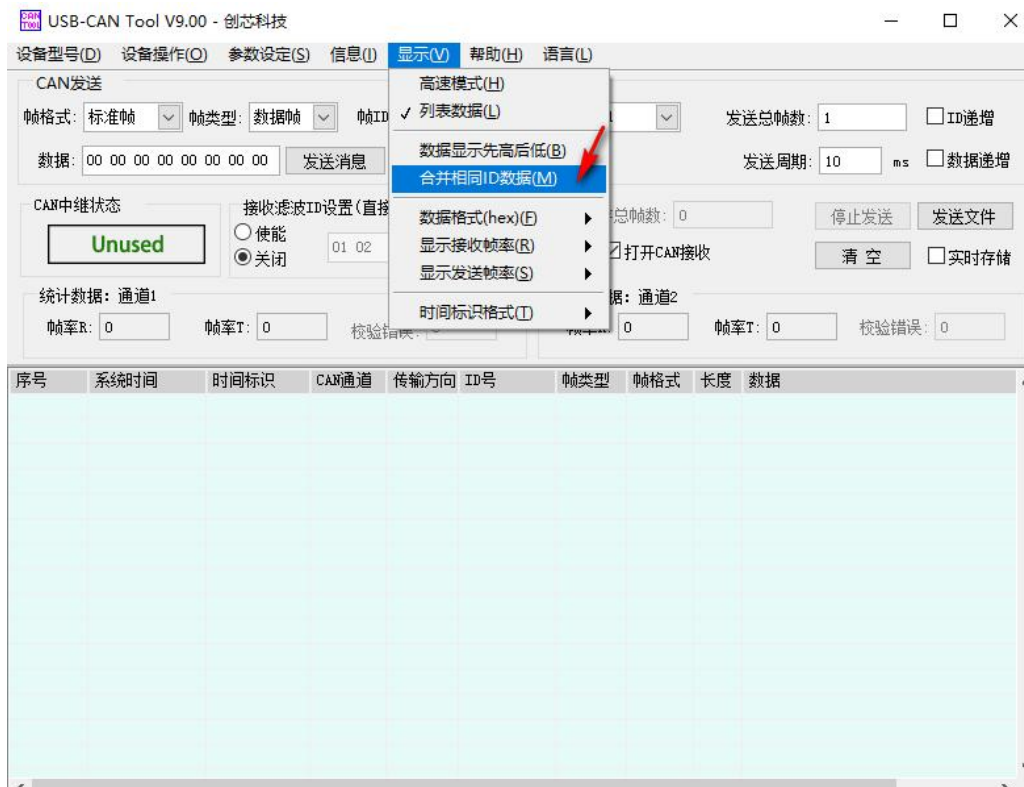
This document introduces several practical functions to assist in the analysis of raw data. The general flow of raw data can be quite large, making it difficult to find the corresponding ID and data bits. These functions aim to simplify the data processing and analysis.

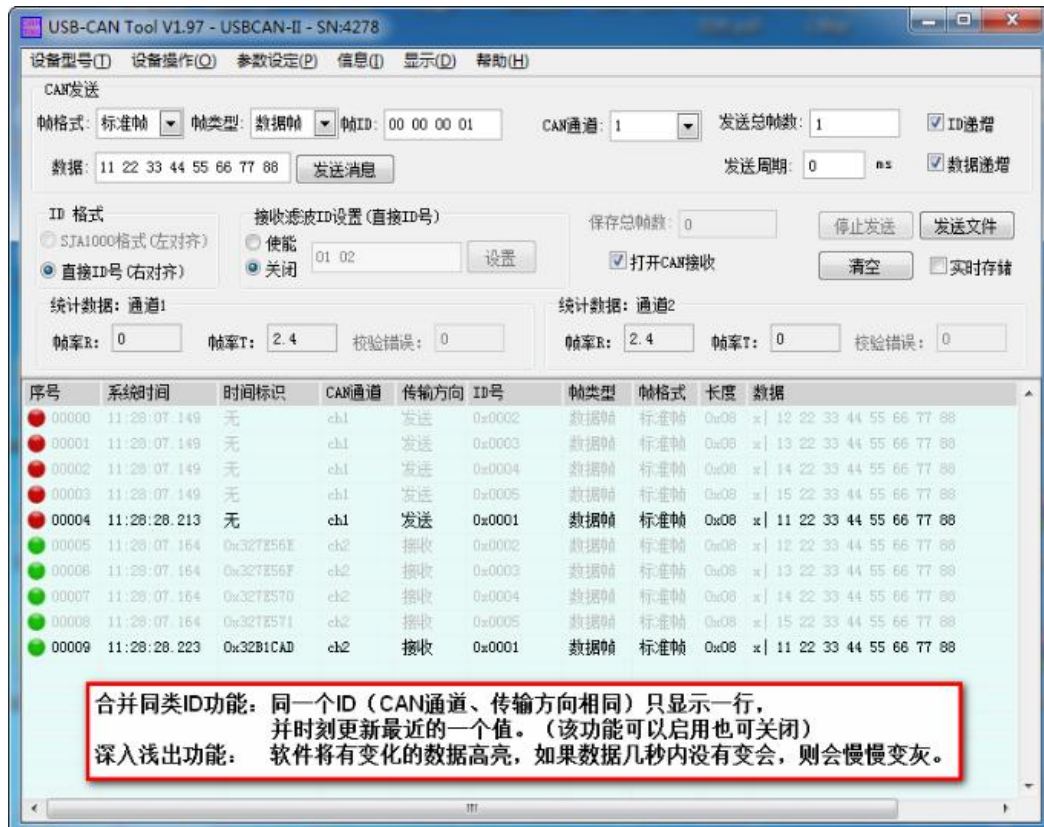
1. The USB_CAN TOOL Is An Original Debugging Tool That Combines the Same ID and In-Depth Features.

To operate the USB_CAN TOOL software, please refer to the '3.USB-CAN Tool Debugging Software Installation and User Manual.pdf'. After receiving the raw CAN data, select the 'Merge Same ID Data' option under the 'Display' list in the menu to activate the 'Merge Same ID' and 'Deeper and Deeper' functions.

Merge similar ID functions. A device typically sends a limited amount of ID data, where the same ID data represents the same meaning. During general debugging, it is only necessary to focus on the current (last) frame of data.

The Insight function highlights data updates for the same ID in black and greys out the line where the ID is located if it has not changed within 5 seconds. An update refers to data with the same ID appearing on top of the bus, regardless of whether the data bits have changed or not. This function is useful for cracking unknown protocols and can assist the user in quickly identifying variables.





2. Chou-Li-Gong CANTest, CANPro Software Change Data "Red" Function

To use CANTest or CANPro1.50, simply replace ControlCAN.dll and other library files (refer to: How to use Zhou Ligong CANTest/CANpro1.50 software compatibly.pdf) and select the USBCAN-2E-U model.

CANpro1.50 has rich functions, here take CANpro1.50 as an example.

To begin, launch the CANpro1.50 software and choose the USBCAN-2E-U interface card. Select the appropriate baud rate for the bus (typically 500K for automotive CAN bus), then click OK to start the interface card. Refer to Figure 2 for guidance.

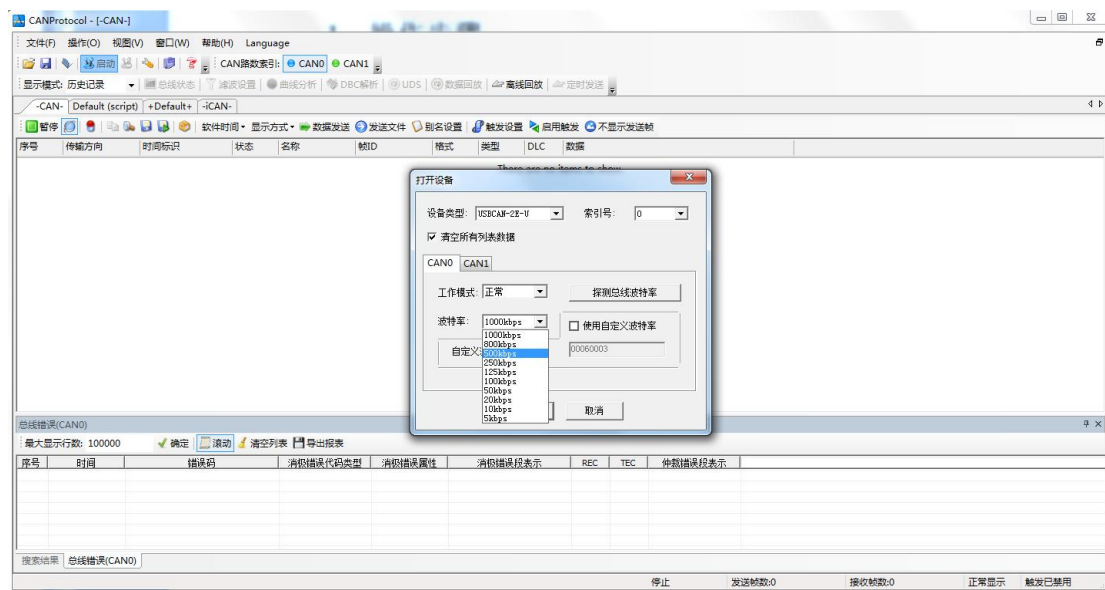


Fig. 2 Opening CANpro software initialisation

Click the DBC Parsing button in the menu shortcut operations to enter the DBC Parsing interface, as shown in Figure 3;

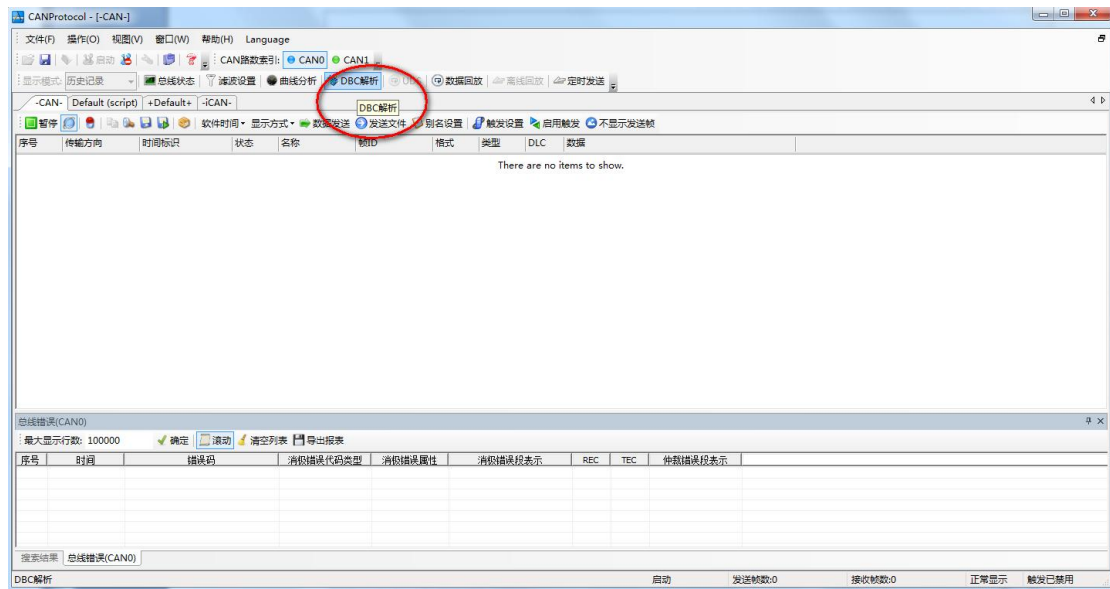


Figure 3 Open DBC Parsing

When using the classification display function in the DBC parsing interface, the software will highlight data with changes in red. This can help users quickly identify variables when cracking unknown protocols. For instance, to determine the CANID and data segment corresponding to the steering wheel, users can turn the steering wheel and observe the variables that turn red.

The 'red marking' should allow for the 'sorting display' function. The marking is based on the unit of byte, and any change in a bit within the byte will result in the entire byte being marked as 'red'. The marking is byte-based, meaning that any change in a bit within the byte will result in the entire byte being marked as 'red'.

序号	传输方向	时间	消息名	ID	源地址	目的地址	数据类型	帧格式	数据长度	帧数据
0	接收	532.2626	EEC2	00F00302 H	02 H	--	扩展帧	数据帧	8	00 00 30 00 00 00 00 00
1	接收	532.2489	EEC1	0CF0041A H	3A H	--	扩展帧	数据帧	8	00 00 00 60 D6 00 00 00
2	接收	532.2598	HOURS	00FEE505 H	05 H	--	扩展帧	数据帧	8	D2 09 00 00 00 00 00 00
3	接收	532.2070	ET1	00FEE01 H	01 H	--	扩展帧	数据帧	8	14 14 00 00 00 00 00 00
4	接收	532.2215	VEP1	00FEF704 H	04 H	--	扩展帧	数据帧	8	00 00 00 00 00 00 D0 68
5	接收	532.2267	SHUTDN	00FEE407 H	07 H	--	扩展帧	数据帧	8	00 00 00 00 00 00 00 00
6	接收	532.2422	EFL_P1	00FEEF03 H	03 H	--	扩展帧	数据帧	8	00 00 00 27 00 00 00 00
7	接收	532.2458	IC1	00FEF606 H	06 H	--	扩展帧	数据帧	8	00 00 27 00 00 00 00 00

Fig. 5 DBC protocol parsing result

3、Application Scope

- Industrial control testing
- Automotive electronics maintenance and repair
- Protocol cracking